PROJECT REPORT

Event Force Management System

A Salesforce CRM solution for Event Planner Solutions

TEAM MEMBERS

Santhoshini S (Team Lead)

Priyasaki J

Vaishnavi S

Vivegha P

GITHUB REPOSITORY

github.com/Santhoshini2006/EVENT-MANAGEMENT-SYSTEM

Salesforce Developer project | Report date: 3 October 2026

Event Force Management System: Project Report

Salesforce Developer project | Team lead: Santhoshini S | Team members: Priyasaki J, Vaishnavi S, Vivegha P | GitHub: Santhoshini2006/EVENT-MANAGEMENT-SYSTEM

1. Introduction

Overview. Event Force Management System is a Salesforce CRM solution built for Event Planner Solutions. It digitizes and automates event management: client bookings, venue reservations, vendor coordination, cancellation approvals, client reminders and feedback collection. Coordinators, vendor managers, clients and admins work in one Lightning app, Event Planner, on the same centralized data.

Purpose.

- Centralize event, client, vendor, venue and feedback data in one CRM platform
- Simplify event booking and prevent double bookings of a venue on the same date
- Automate reminders, cancellation approvals and event completion; calculate the budget from the event type
- Provide role-based access and real-time visibility through reports and dashboards

Business targets. The requirement analysis set targets of about 60% fewer manual processes and about 40% better booking efficiency. These are planning targets and were not measured in this prototype.

2. Ideation Phase

2.1 Problem statements

PS	I am	I'm trying to	But	Because	Feel
PS-1	Client	book an event and get updates	booking depends on calls	details are not centralized	frustrated
PS-2	Event Coordinator	schedule events, manage approvals	everything is manual	no automated workflow	overloaded
PS-3	Vendor Manager	see which events need my vendors	assignments are scattered	no record links events and vendors	confused
PS-4	Client	get a reminder before my event	reminders depend on staff	communication is not automated	uncertain
PS-5	Event Admin	monitor events, control access	data is spread across files	no central reporting or security model	concerned

2.2 Empathy map (Client)

Think & feel: wants venue, vendors and date confirmed. **Hear:** confirmations, reminders and cancellation notices. **See:** event name, date, type, venue, status. **Say & do:** requests services, asks for updates, gives feedback. **Pain:** double bookings, missed reminders, unclear cancellation status. **Gain:** timely reminders, clear status, one place for feedback.

2.3 Brainstorming

Ideas considered: paper registers, spreadsheets, a standalone event portal and Salesforce CRM. Selected: Salesforce CRM for centralized data, Flow and Apex automation, approval processes, role-based security and built-in reports.

3. Requirement Analysis

3.1 Client journey

Event Request > Event Record > Venue Reservation > Vendor Assignment > Confirmation > 3-Day Reminder Email > Event Day > Completion (or Cancellation Approval) > Feedback

3.2 Functional requirements

FR	Requirement	Implemented by
FR-1	Event booking, record creation	Event__c object, Event Planner app
FR-2	Event status tracking	Event_Status__c (Planned, Confirmed, Completed, Pending Cancellation, Canceled, Rejected)
FR-3	Automatic budget	Event_Budget__c formula (CASE on Event Type)
FR-4	Client info, valid email	Client__c, validation rule Email_Valid_Address
FR-5	Vendor management	Vendor__c, Service_Type__c multi-select picklist
FR-6	Venue management	Venue__c, Availability_Status__c, VenueStatusHelper class
FR-7	Prevent double booking	PreventDoubleBooking Apex trigger
FR-8	Event-to-vendor link	EventVendor__c junction object
FR-9	Cancellation approval	Event_Cancellation_Process, email templates, field updates
FR-10	Client reminder	Flow: Client Reminder - 3 Days Before
FR-11	Auto-complete past events	BatchCompleteEvents, ScheduleCompleteEvents (daily 8 PM)
FR-12	Feedback collection	Feedback__c (auto number F-{0000}), Event and Client lookups
FR-13	Role-based access	Profiles, roles, Feedback Manager permission set, OWD
FR-14	Reports and dashboards	Upcoming Events by Month report, Event Force Operations Dashboard

Non-functional: usability (Lightning app with tabs for every object), security (profiles, roles, permission set, OWD), reliability (validation rules and triggers), performance (Batch Apex in scopes of 200), scalability (batch Apex, lookup and junction relationships), maintainability (declarative flows and approvals, small helper classes).

3.3 Data flow

Client > Event Request > Event Record > Venue + Client lookups > Vendor assignment (Event Vendor) > Status update > Reminder / Approval > Completion > Feedback > Reports & Dashboards. Automation: status Confirmed > scheduled path 3 days before Event Date > reminder email to the client. Status Pending Cancellation > approval process > Canceled or Rejected > email to the event owner.

3.4 Technology stack

Layer	Technology
Platform / UI	Salesforce Developer Edition; Lightning App (Event Planner) with custom object tabs
Database / Logic	Custom objects; record-triggered Flow, Apex triggers, Batch and Schedulable Apex
Approvals / Notifications	Approval process with classic email templates; email alerts and flow email action
Security / Reporting	Profiles, roles, permission sets, sharing settings; reports and dashboards

4. Project Design

4.1 Problem-solution fit

Problem	Solution	Benefit
Manual event booking	Event management in Salesforce	Centralized booking
Venue clashes	Trigger blocks a second event on the same venue and date	No double bookings
Untracked cancellations	Approval process with email alerts	Controlled, auditable cancellations
Missed reminders	Scheduled flow, 3 days before the event	Timely client communication
Scattered data / hard to monitor	Six objects plus reports and dashboards	Structured data, better visibility
Different access needs	Profiles, roles, permission set, sharing settings	Secure access

4.2 Proposed solution

A Salesforce event management system with automated reminders, approval-based cancellations and venue protection through Flow and Apex, giving centralized booking, timely notifications and clear reporting. New objects, flows and roles can be added without redesign.

4.3 Data model

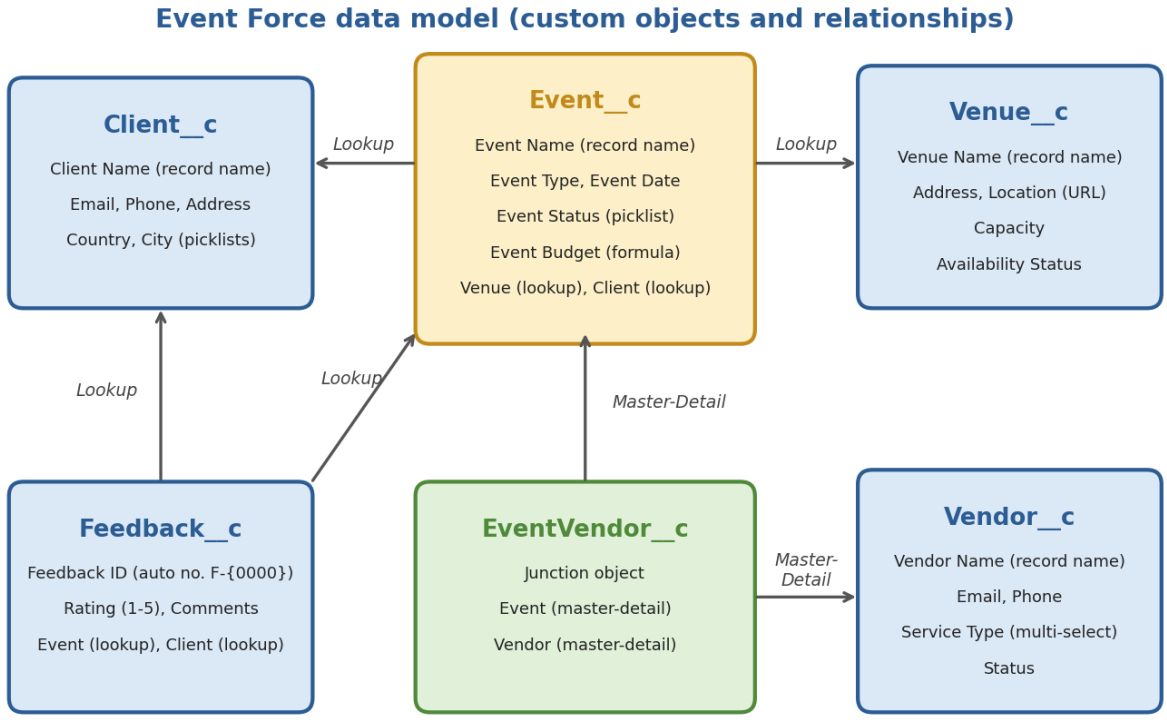


Diagram 1: Event Force data model

4.4 Architecture

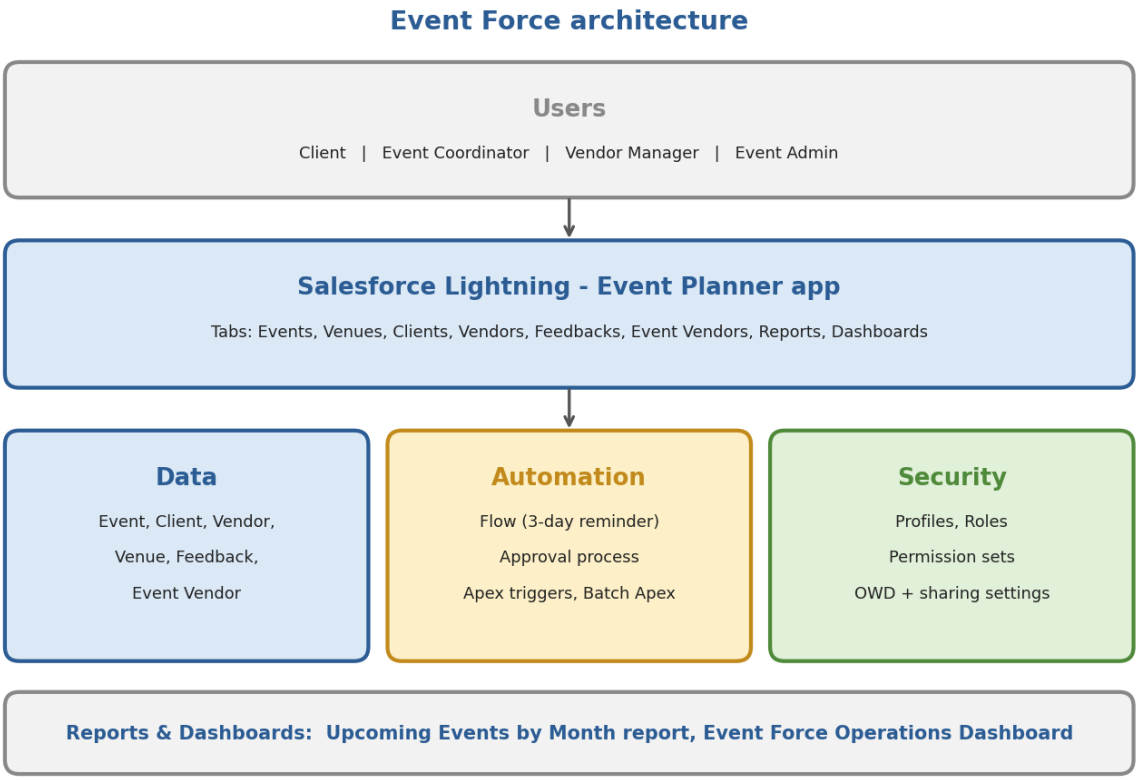


Diagram 2: Event Force architecture

5. Project Planning

Phase	Milestones	Key activities	Status
1: Requirement analysis & planning	Planning	Functional scope, stakeholder mapping, roadmap	Completed
2: Backend development	1 to 10	Org, objects, tabs, fields, relationships, validation, approval, flow, Apex, batch jobs	Completed
3: UI/UX customization	11 to 13	Event Planner app, report, dashboard	Completed
4: Data, testing & security	14 to 18	Sample data, profiles, roles, users, permission set, OWD	Completed
5: Deployment & maintenance	Wrap-up	Deployment approach, maintenance plan, report, demo video	Completed

Team

Name	Role
Santhoshini S	Team Lead
Priyasaki J	Team Member
Vaishnavi S	Team Member
Vivegha P	Team Member

6. Development Phases

- **Phase 1 - Requirement analysis:** business objective, scope (six custom objects, relationships, flows, validation, approval, Apex, security, reports), five stakeholder groups, seven-step roadmap.
- **Phase 2 - Backend:** Developer Edition org; Event, Client, Vendor, Venue, Feedback and Event Vendor objects; fields incl. Event Type, Event Date, Event Status and the Event Budget formula; Lookup and Master-Detail relationships; email validation rule; cancellation approval with email templates; 3-day

reminder flow; VenueStatusHelper with EventTrigger13; PreventDoubleBooking trigger; BatchCompleteEvents and ScheduleCompleteEvents (daily 8 PM).

- **Phase 3 - UI/UX:** Event Planner Lightning app with all tabs, report folder, Upcoming Events by Month report and Event Force Operations Dashboard.
- **Phase 4 - Data and security:** sample records for every object; Event Admin, Event Coordinator, Vendor Manager and Client profiles and roles; users; Feedback Manager permission set; organization-wide defaults for Client, Event, Feedback, Vendor and Venue set to Public Read/Write internally and Private externally.
- **Phase 5 - Deployment:** built and tested in a Developer Edition org. Production deployment is described but not performed (real deployments use sandboxes, change sets or Salesforce DX). Maintenance covers monitoring flows, Apex jobs and approvals, and fixing access issues.

7. Functional and Performance Testing

"Configured" means the feature was set up as documented but no screenshot is included. Figure numbers refer to Section 12.

Area	What was verified	Expected result	Evidence
Objects and tabs	Custom objects and tabs	Visible in Object Manager and the app	Fig 1 to Fig 5, Fig 13
Event fields	Picklists and Event Budget formula	Budget returned by Event Type	Fig 6
Record creation	New Event with venue and client lookups	Record saved and listed	Fig 22, Fig 23
Reminder flow	Client Reminder - 3 Days Before	Email 3 days before Event Date	Fig 7 to Fig 10
Scheduled Apex	Batch and Schedulable classes	Past events marked Completed daily	Fig 11
Dashboard	Event Force Operations Dashboard	Charts show upcoming events	Fig 29
Security	Profiles, roles, permission set, OWD	Access follows role	Fig 14 to Fig 21
Related data	Venues, vendors, clients, feedback, event vendors	Sample records visible in the app	Fig 24 to Fig 28
Approval, validation rule, triggers, report	Configured as documented	Cancellation approved; invalid email blocked; double booking blocked	Configured

8. Advantages and Disadvantages

Advantages: centralized data, automated reminders, controlled cancellations, no double bookings, automatic budget, role-based security, clear reporting, scalable on Salesforce.

Disadvantages: needs accurate data entry and correct configuration, flows and Apex need upkeep, the Event Budget uses fixed amounts per type, and users need the right access.

9. Conclusion

Event Force shows how Salesforce objects, Flow, approval processes, Apex and analytics combine into one event management solution. Coordinators, vendors and clients work from the same data, and reminders, cancellations, venue protection and completion run automatically. Built in a Developer Org for academic purposes, it follows real-world CRM practice.

10. Future Scope

Experience Cloud client portal, vendor performance and feedback reports, budget values from a custom setting, payment and invoice tracking, calendar and messaging integration, sandbox testing and CI/CD.

11. Appendix

A. Objects: Event, Client, Vendor, Venue, Feedback, Event Vendor | B. Automation: flow, approval process, validation rule | C. Apex: VenueStatusHelper, EventTrigger13, PreventDoubleBooking, BatchCompleteEvents, ScheduleCompleteEvents | D. UI: Event Planner app, tabs | E. Security: profiles, roles, Feedback Manager permission set, OWD | F. Analytics: Upcoming Events by Month report, Event Force Operations Dashboard | G. Source code: https://github.com/Santhoshini2006/EVENT-MANAGEMENT-SYSTEM

12. Screenshots

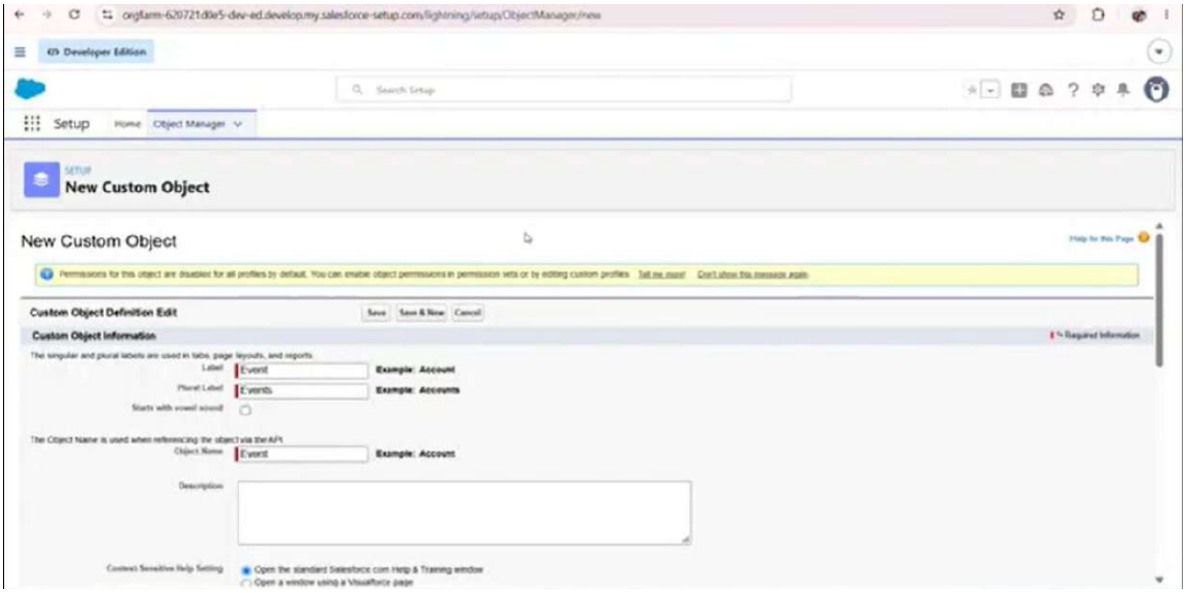


Figure 1: Objects: Event

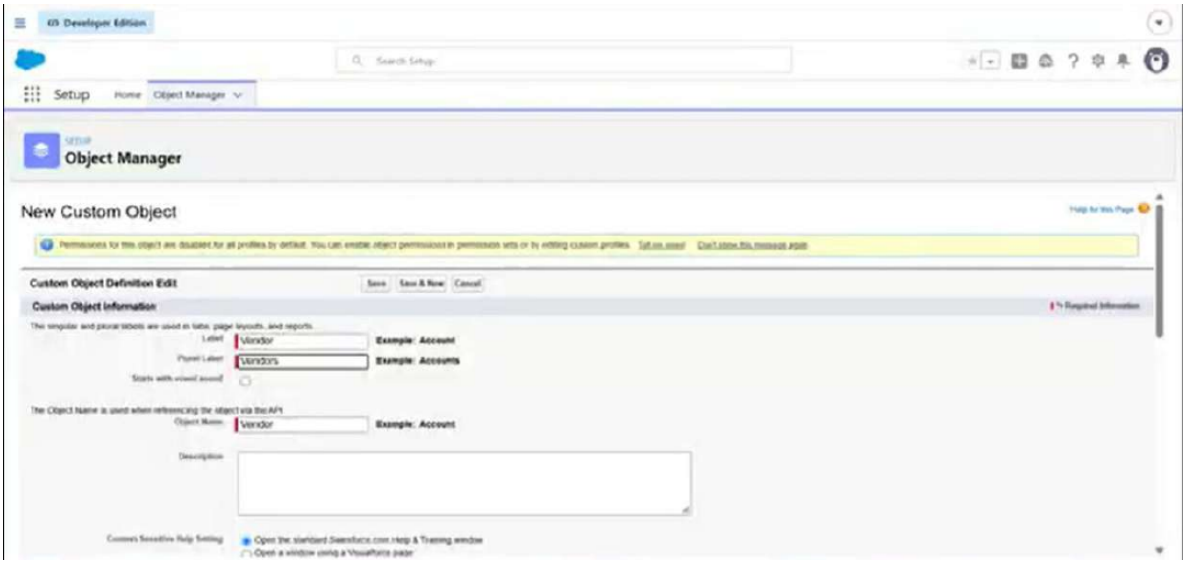


Figure 2: Objects: Vendor

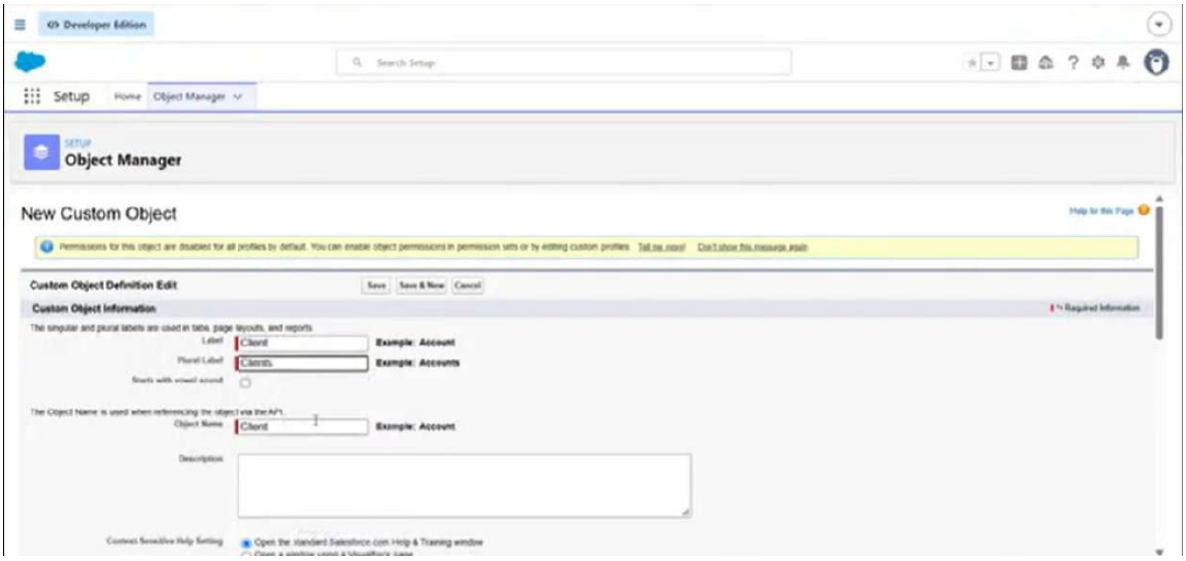


Figure 3: Objects: Client

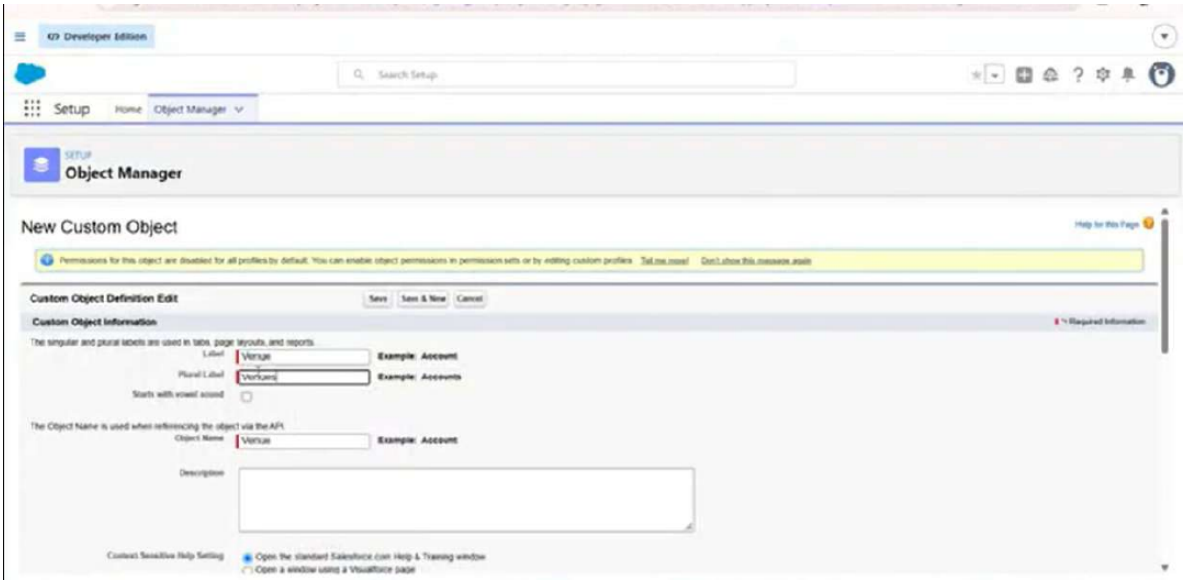


Figure 4: Objects: Venue

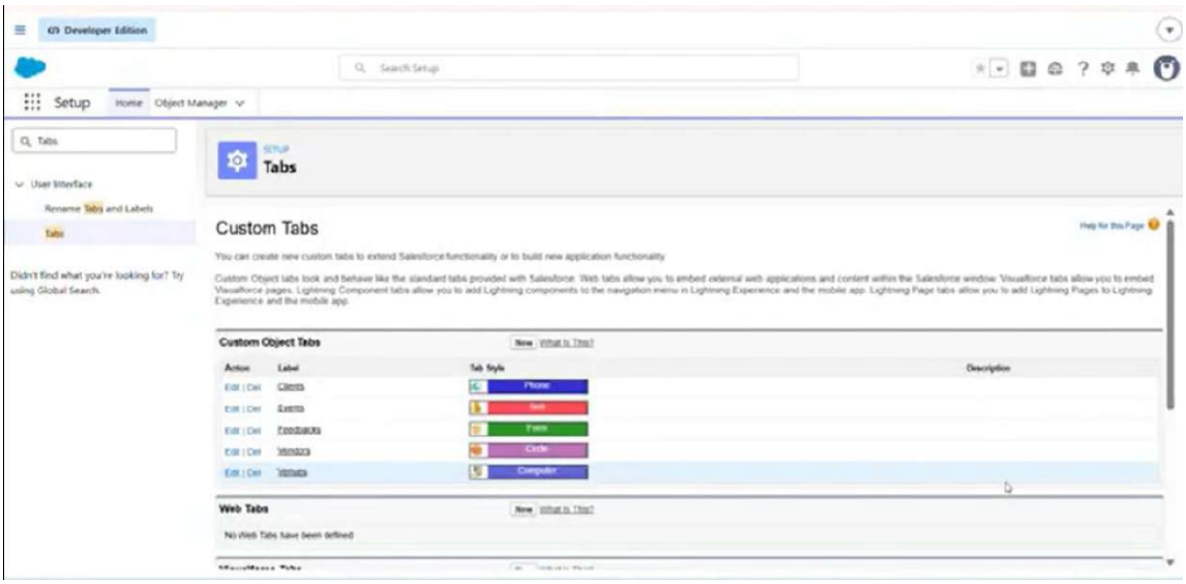


Figure 5: Custom tabs

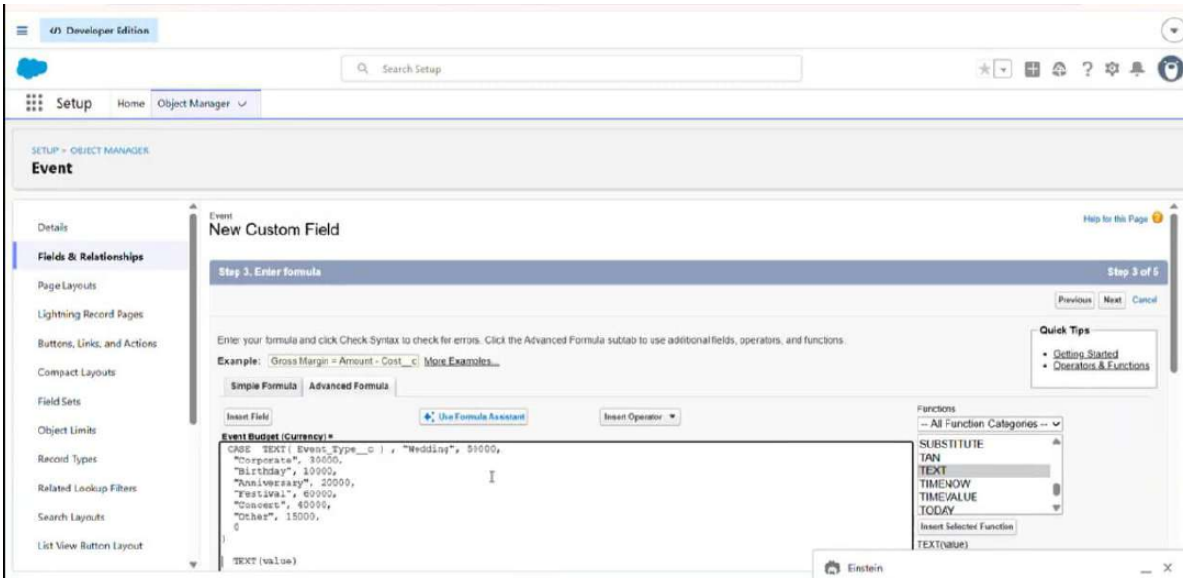


Figure 6: Event Budget formula (CASE on Event Type)

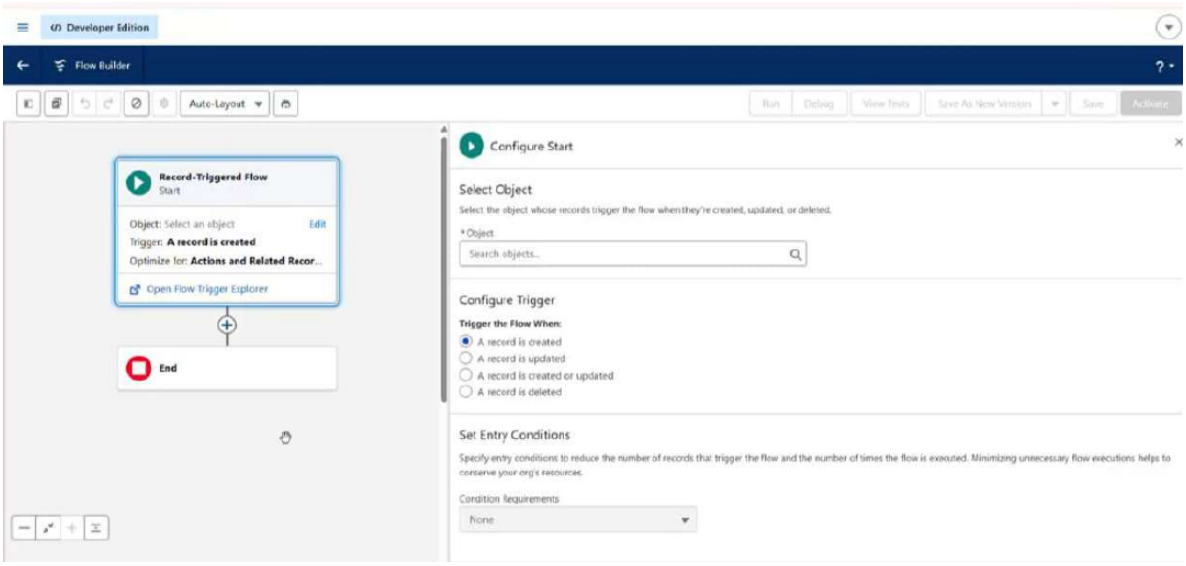


Figure 7: Flow: configure start (record-triggered flow)

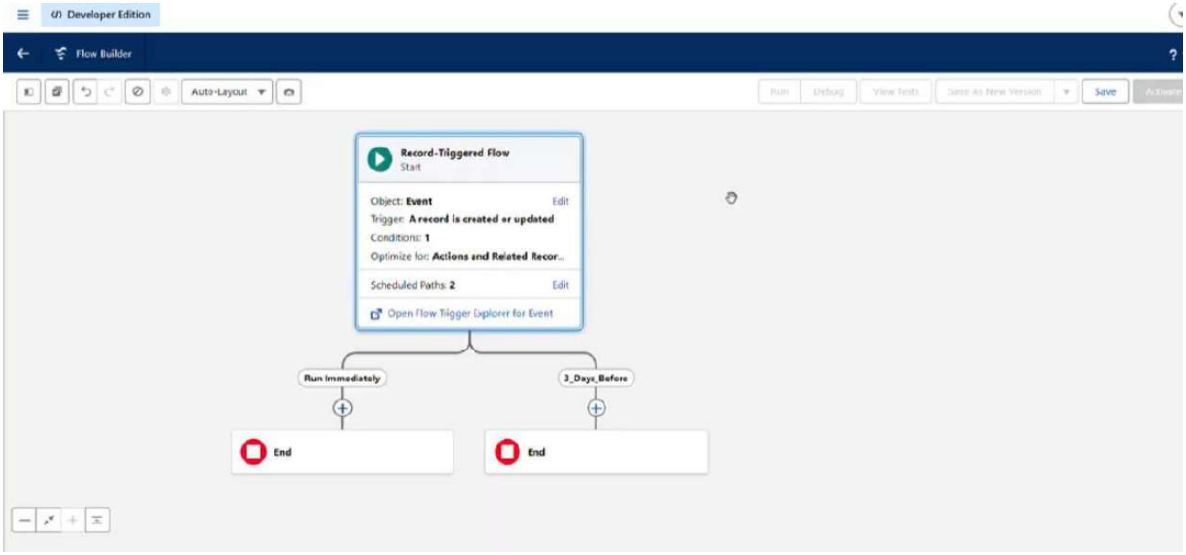


Figure 8: Flow: scheduled paths (Run Immediately and 3_Days_Before)

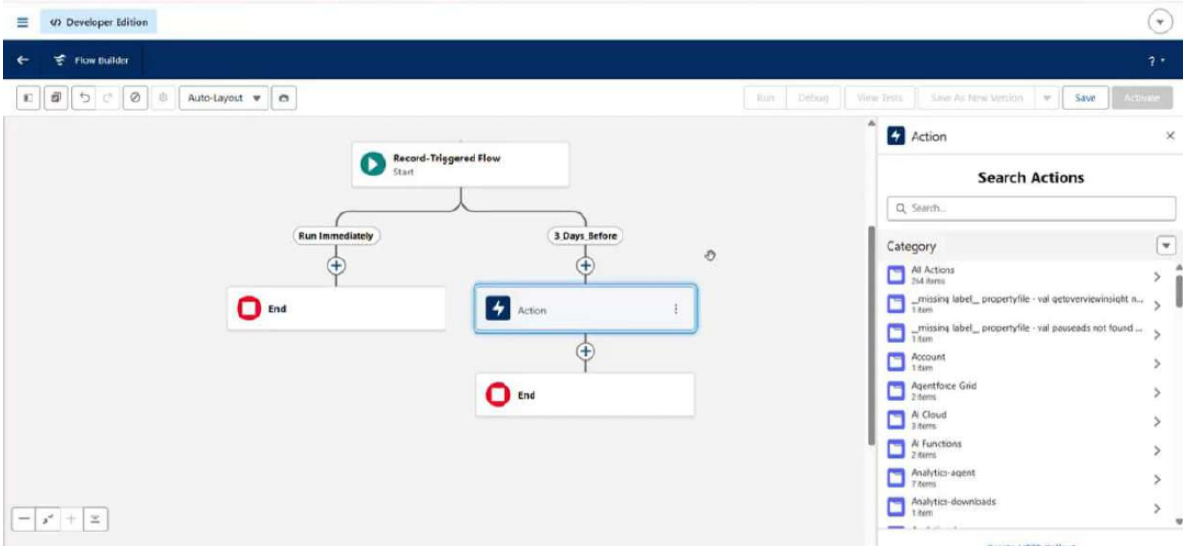


Figure 9: Flow: action added on the 3_Days_Before path

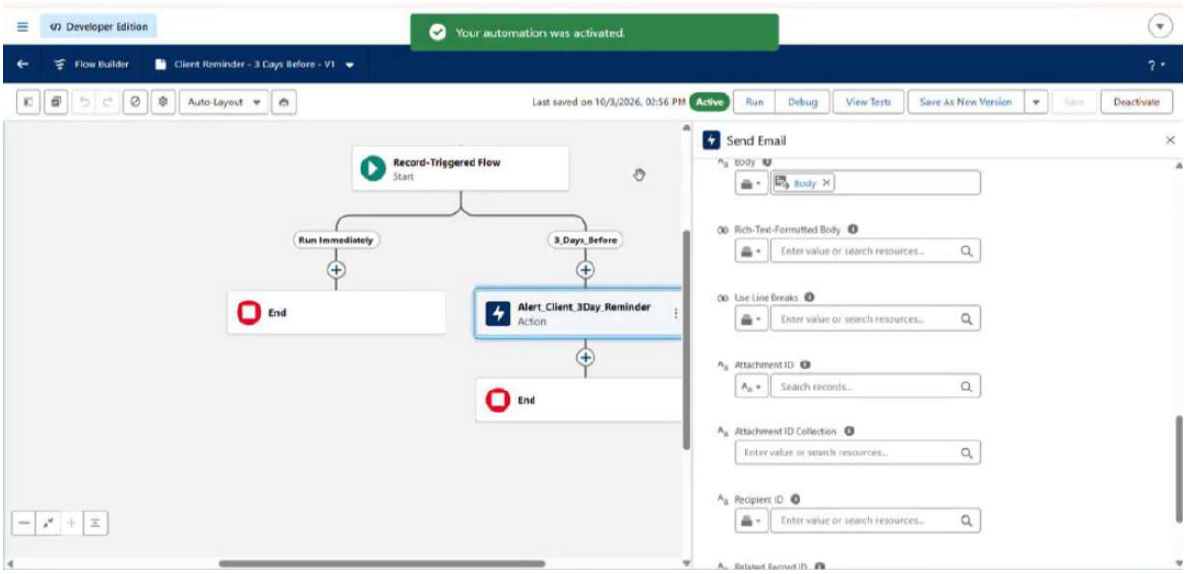


Figure 10: Flow activated with Alert_Client_3Day_Reminder email action

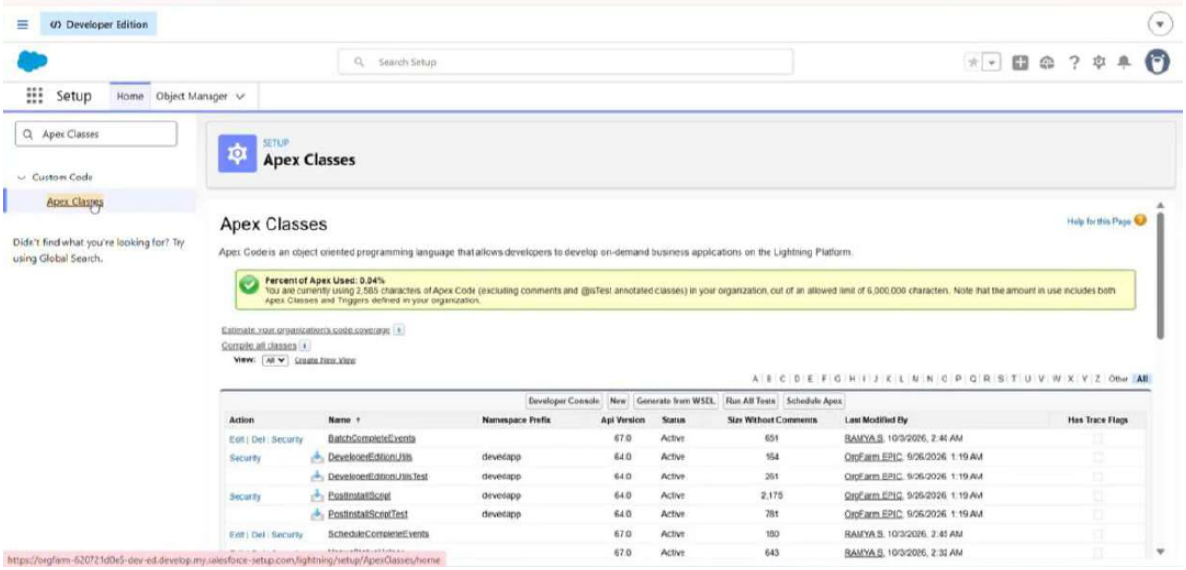


Figure 11: Batch and Schedulable Apex classes

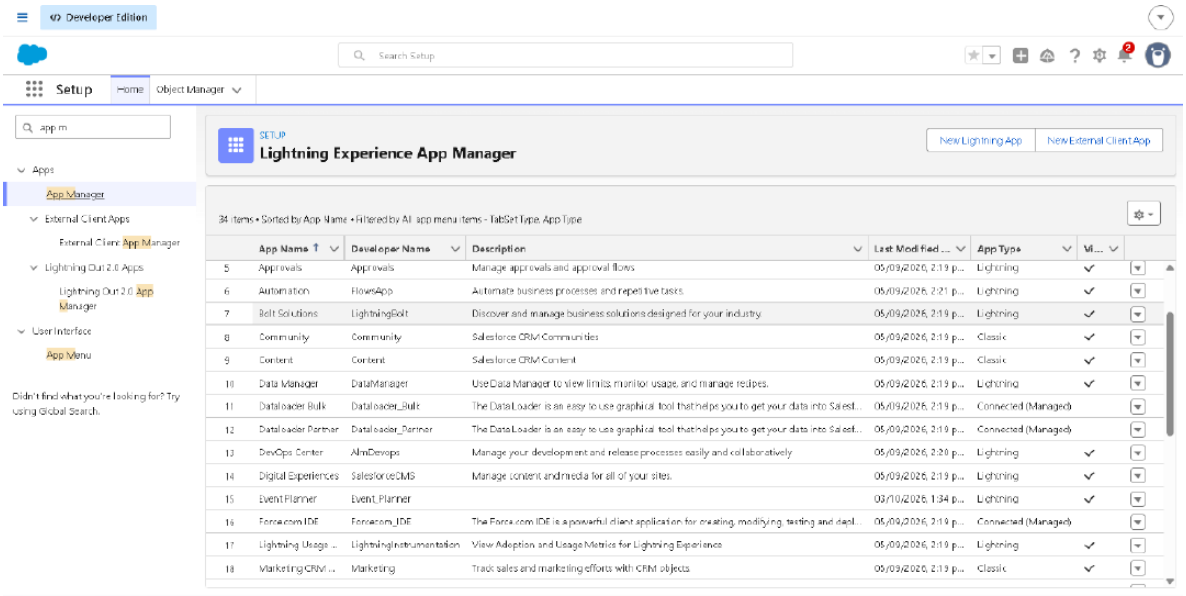


Figure 12: Lightning App Manager: Event Planner app

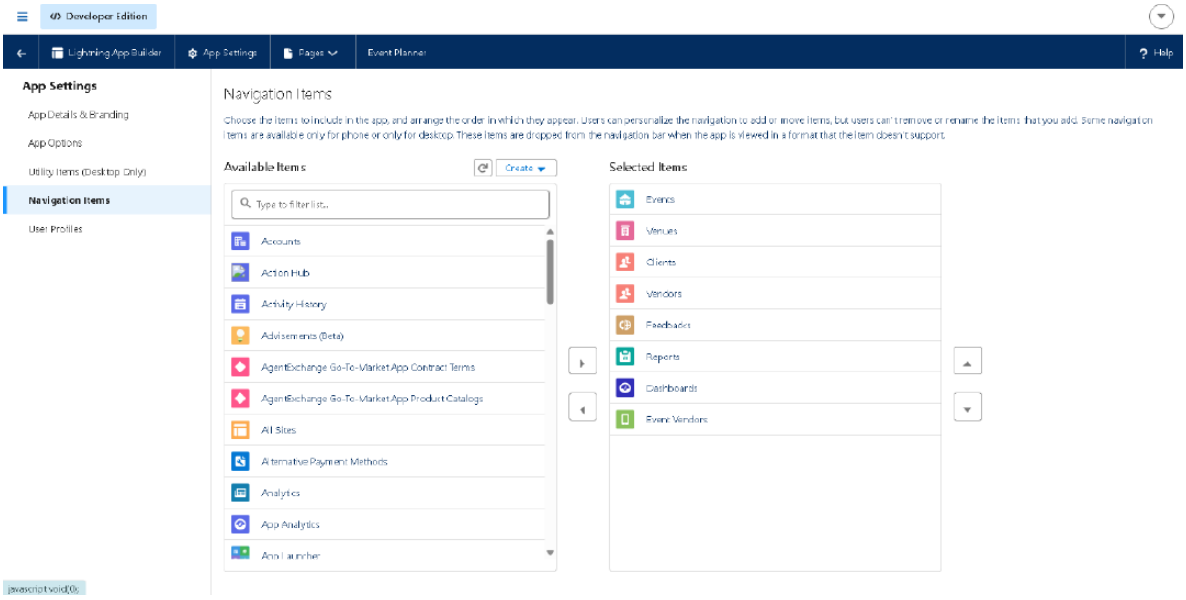


Figure 13: Event Planner app navigation items

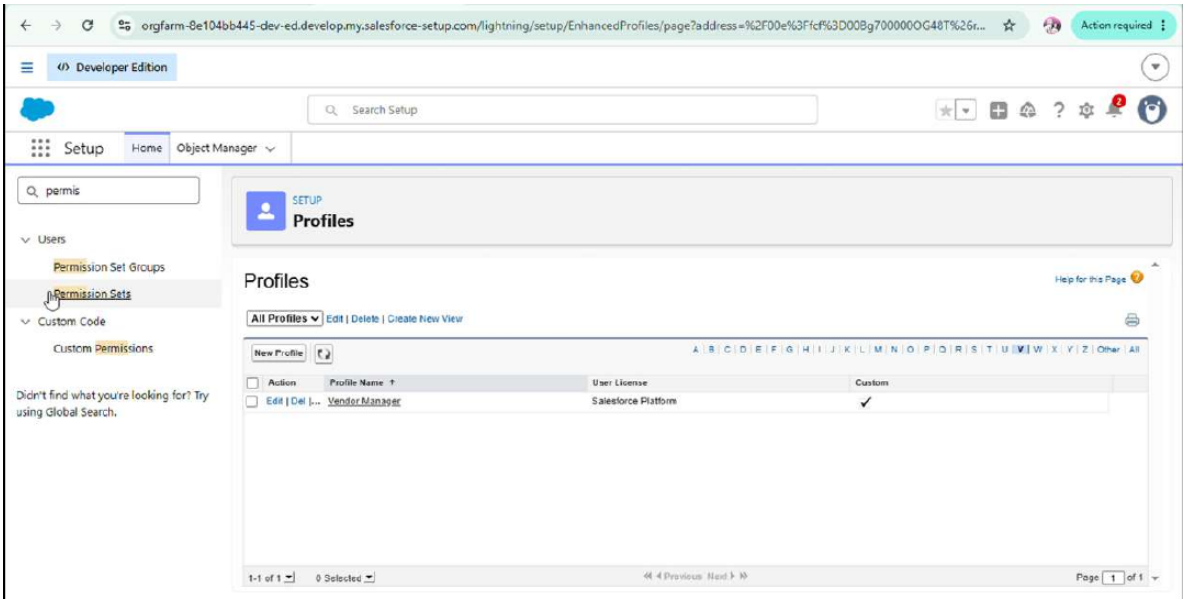


Figure 14: Profile: Vendor Manager

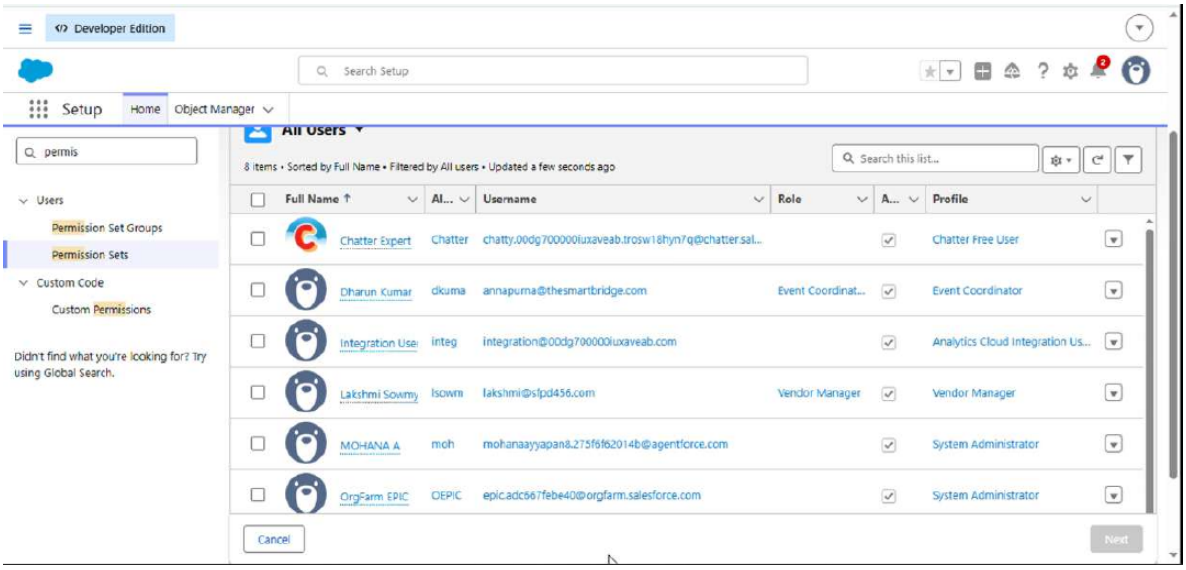


Figure 15: Users with roles and profiles

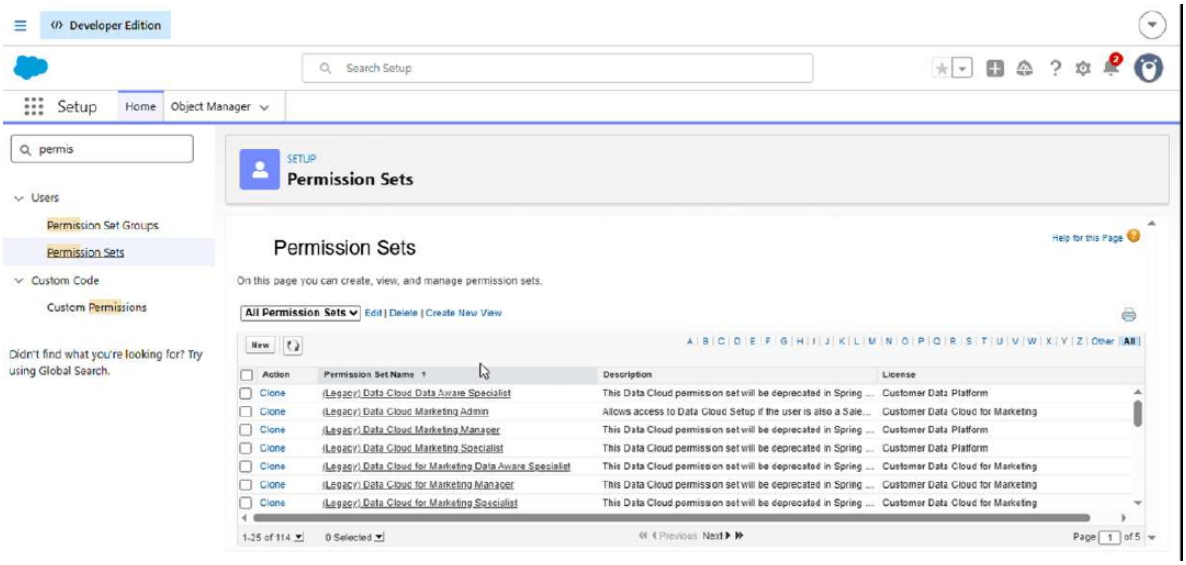


Figure 16: Permission sets list

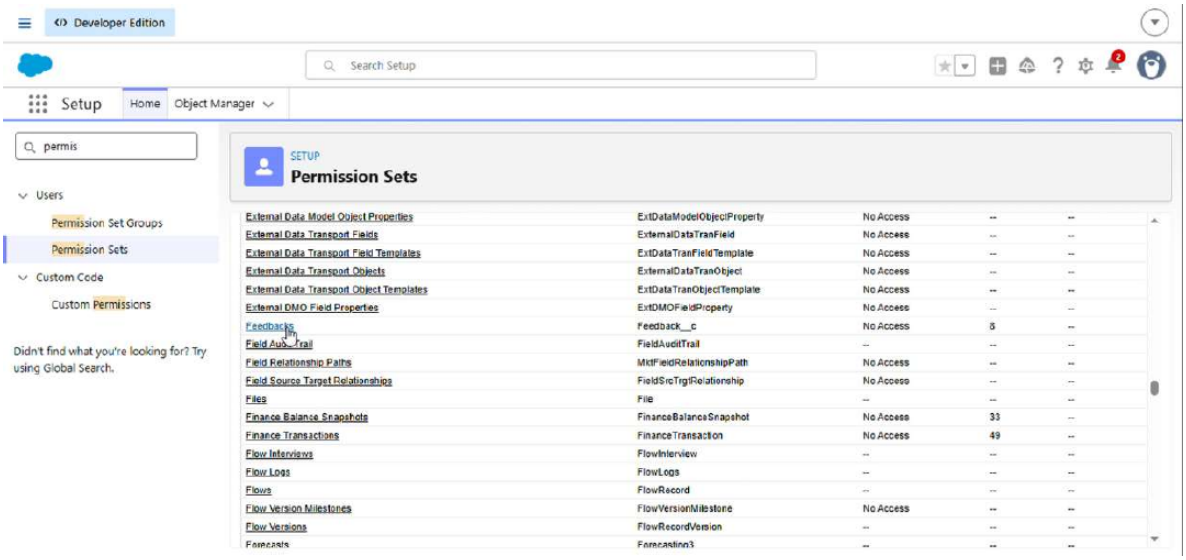


Figure 17: Permission set: object settings for Feedbacks

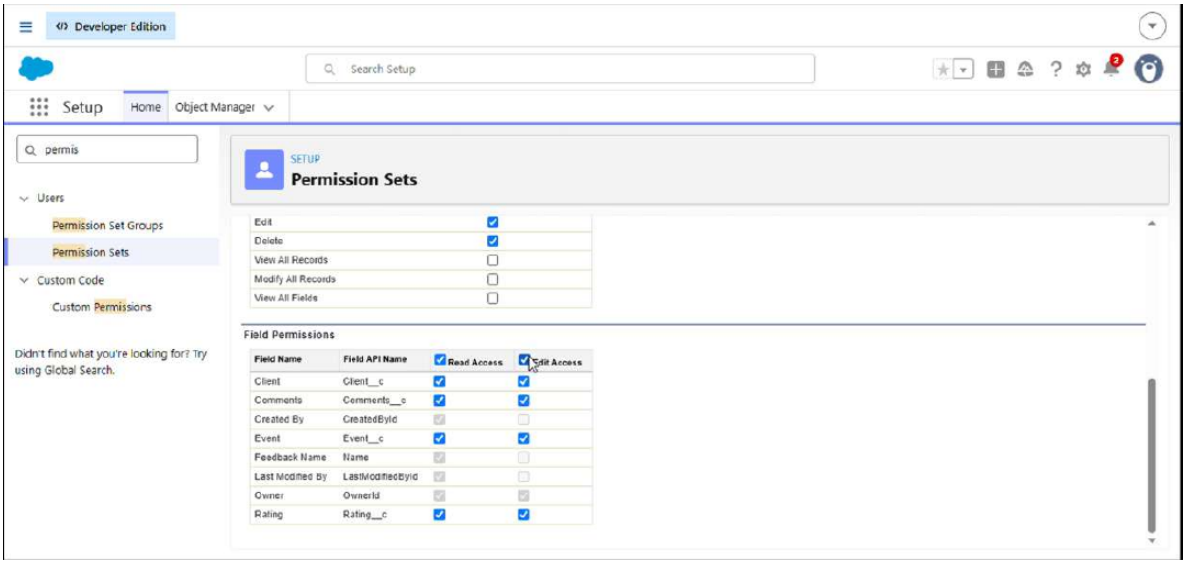


Figure 18: Feedback permission set: field permissions

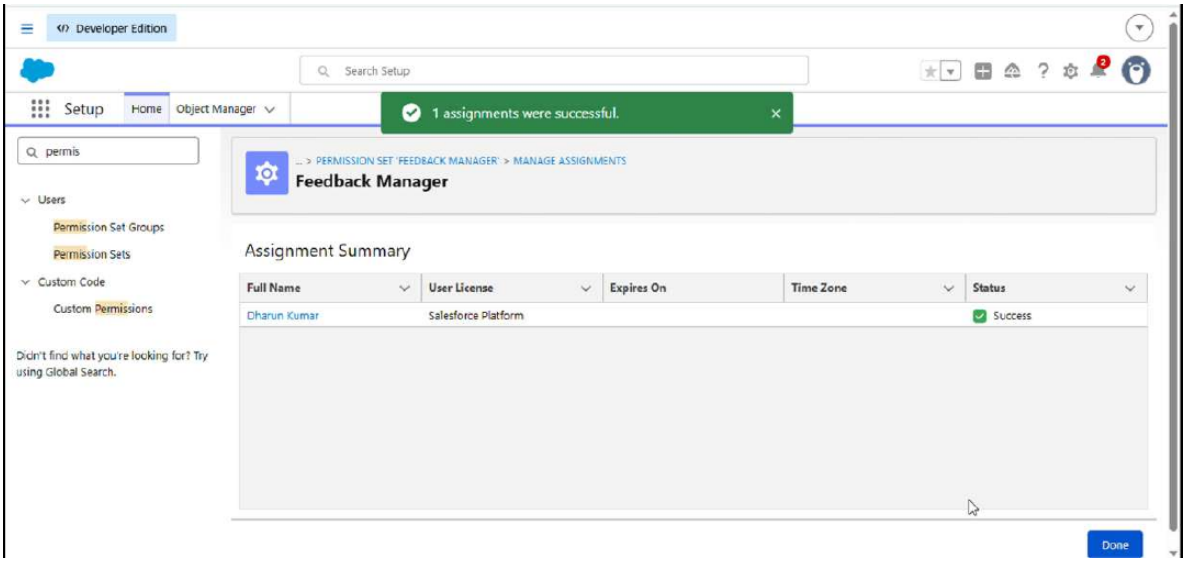


Figure 19: Feedback Manager permission set assignment

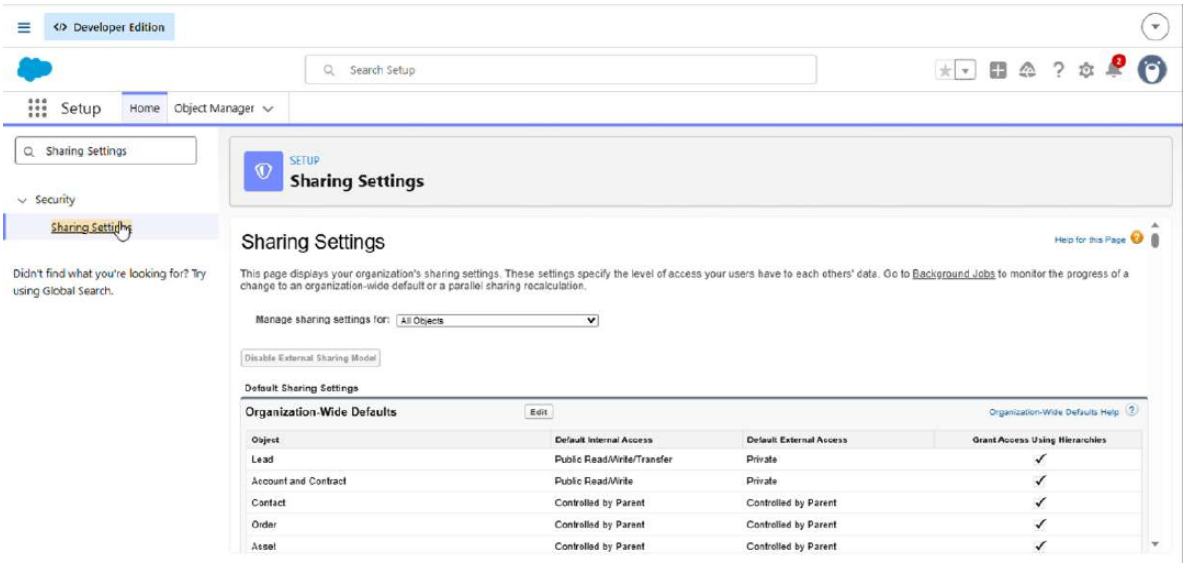


Figure 20: Sharing settings: organization-wide defaults

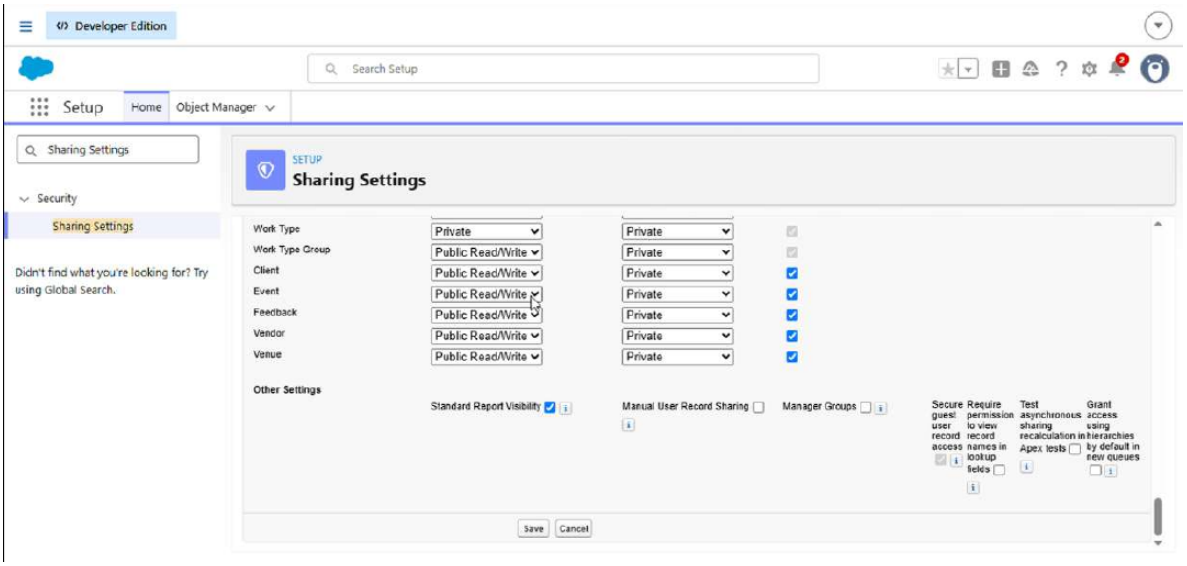


Figure 21: OWD for Client, Event, Feedback, Vendor and Venue

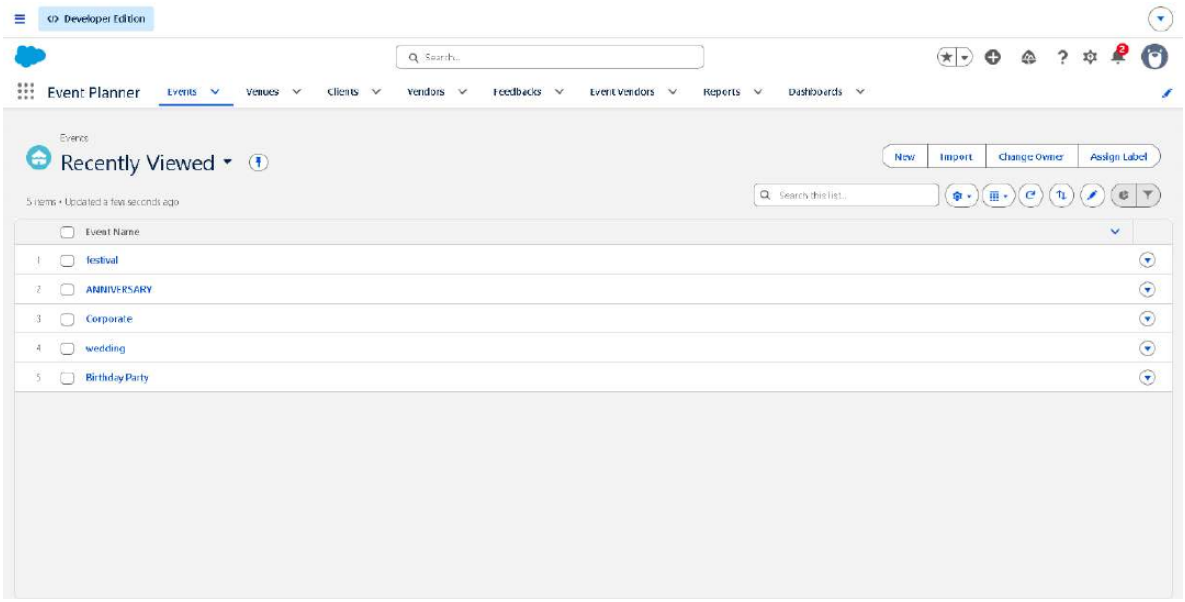


Figure 22: Event Planner app: Events

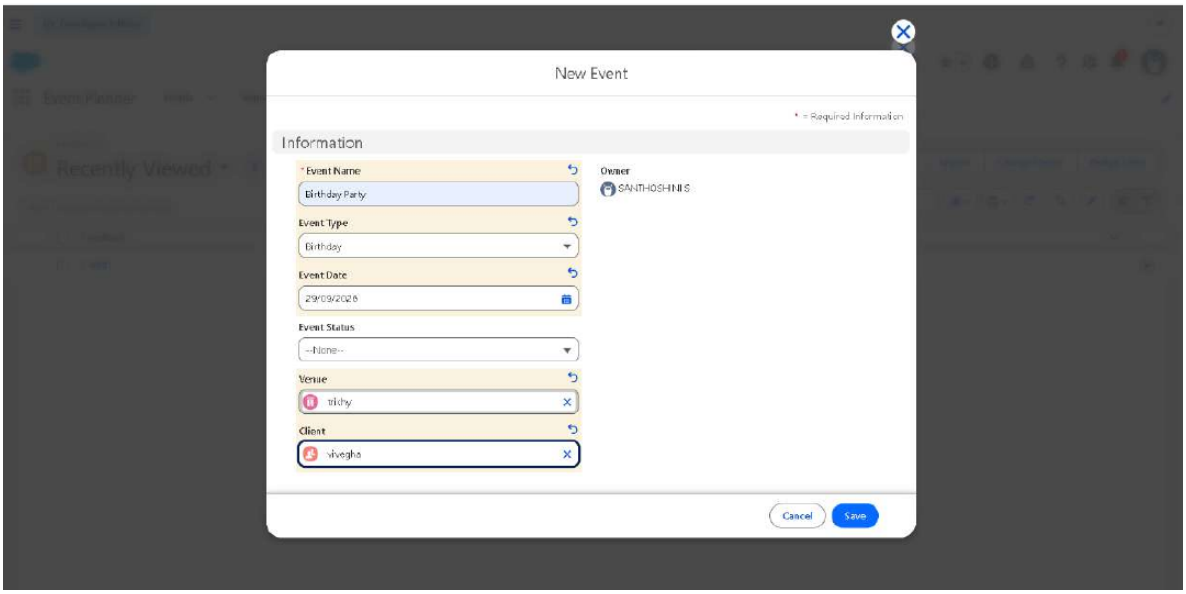


Figure 23: Creating a new Event record

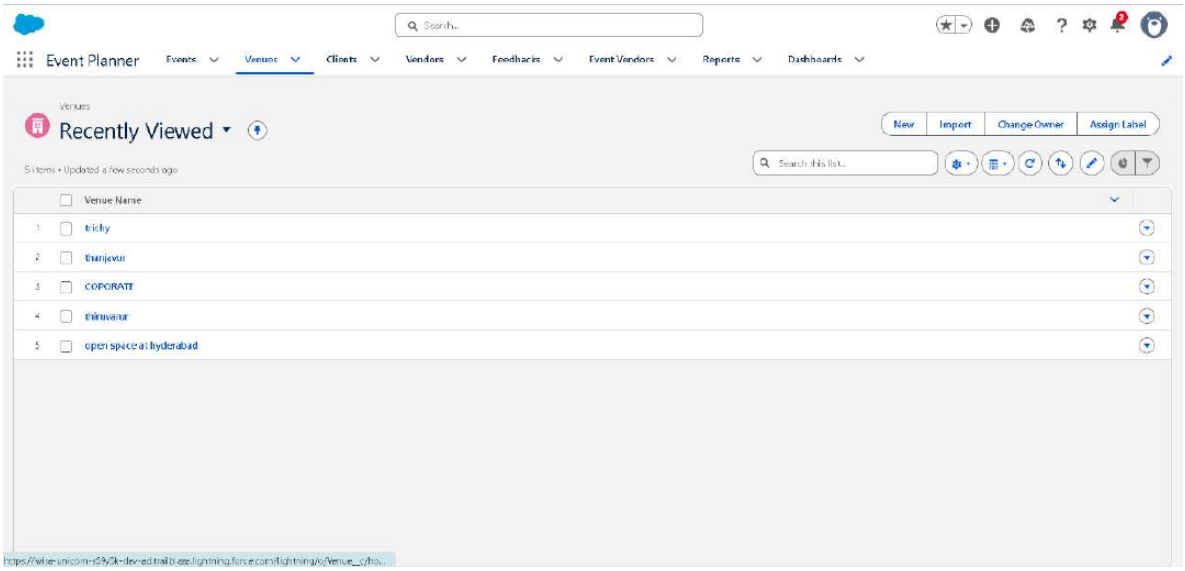


Figure 24: Event Planner app: Venues

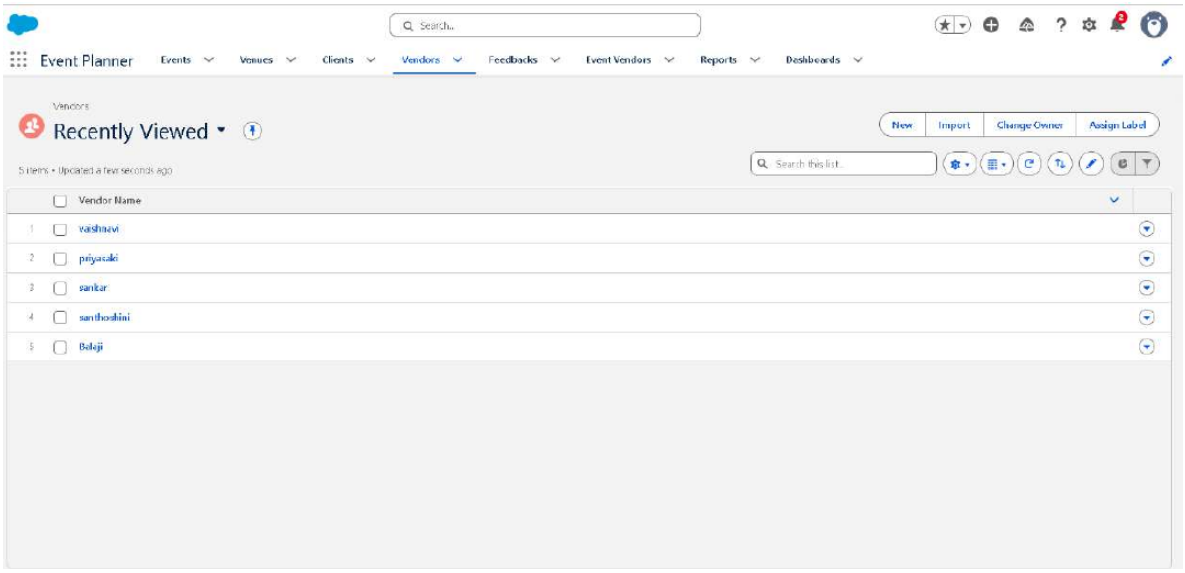


Figure 25: Event Planner app: Vendors

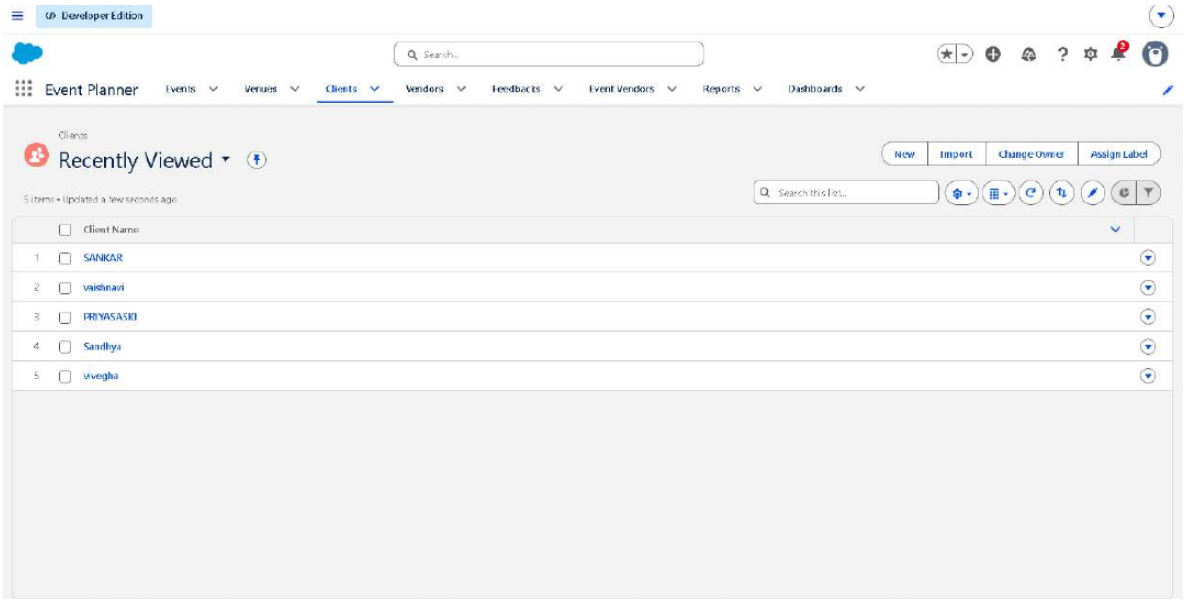


Figure 26: Event Planner app: Clients

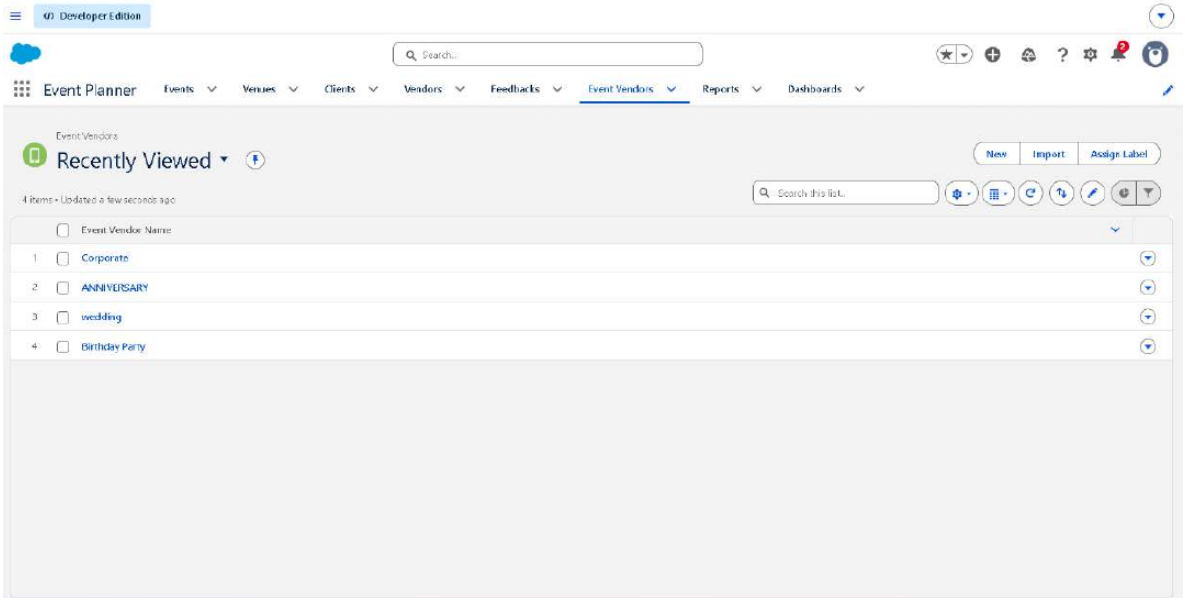


Figure 27: Event Planner app: Event Vendors

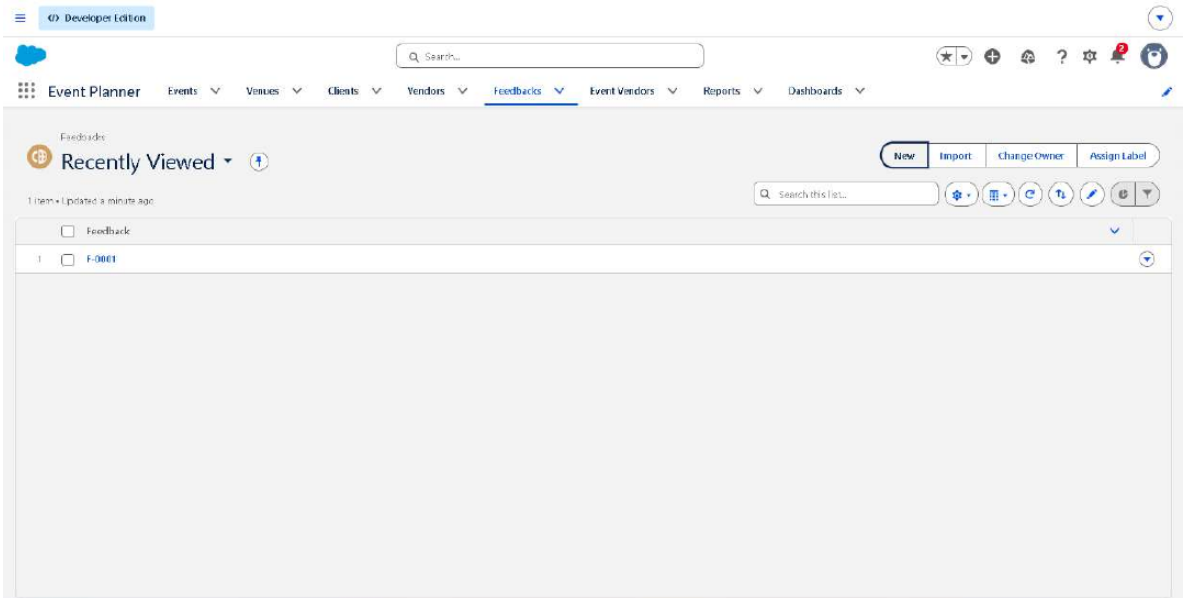


Figure 28: Event Planner app: Feedbacks

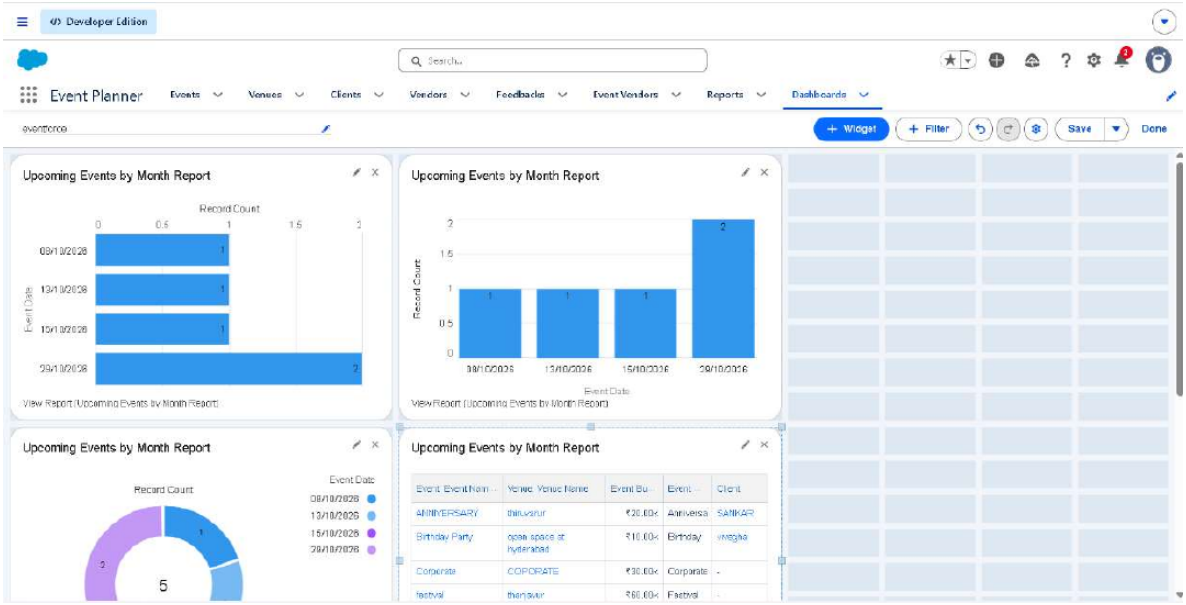


Figure 29: Event Force Operations Dashboard